TRINAY

AI-ASSISTED RETINAL SCREENING REPORT

CONFIDENTIAL MEDICAL DOCUMENT

DATE: 11 Oct 2026

DEPARTMENT OF TELE-OPHTHALMOLOGY & DIABETIC RETINOPATHY SCREENING

1. PATIENT INFORMATION & SCREENING DETAILS

Patient Name:	**Patient ID:**	**Age / Gender:**	**Patient Phone:**
Aviraj abasaheb Murkute	**PX-001**	**20 Yrs / Male**	**9762481967**
Screening Date:	Examined Eye:	Screening Nurse:	Diabetes Duration:
11 Oct 2026	**Right Eye**	**avi**	**5 Years**

2. IMAGE INFORMATION & QUALITY ASSESSMENT

Image Filename: **a7c10ca6c117.png** Image Quality: **Good** Quality Status: **Suitable for Analysis**

3. AI SCREENING RESULT & CLASSIFICATION

No Diabetic Retinopathy (Level 0)

Deep neural ensemble prediction calibrated for primary tele-screening.

CONFIDENCE SCORE

99.9%

Calibrated Model Output

4. PATIENT-SPECIFIC AI CLINICAL SUMMARY

2 retinal lesion types were detected in this screening, including Microaneurysm, Retinal Hemorrhage, encompassing a total quantified lesion area of 64 px. These findings represent retinal abnormalities that may be associated with diabetic retinal vascular disease. Validated neural classifier staging indicates No Diabetic Retinopathy (Level 0) (99.9% confidence). Pixel measurements reflect localized lesion load and must be correlated with clinical slit-lamp findings. In-person ophthalmologist review and dilated biomicroscopy are recommended.

Total detected lesion footprint: 64 px • Severity staging is derived strictly from the calibrated deep neural ensemble and not lesion pixel area alone.

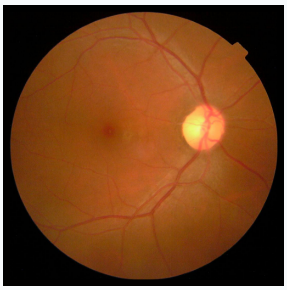
5A. AI-DETECTED FINDINGS (MEASUREMENTS)

Microaneurysm	**Detected (31 px)**
Retinal Hemorrhage	**Detected (33 px)**
Hard Exudate	**Not Detected (0 px)**
Soft Exudate (Cotton-Wool Spot)	**Not Detected (0 px)**

5B. RETINAL ANATOMY LOCALIZATION

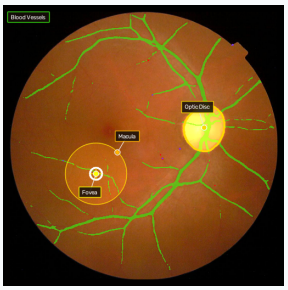
Optic Disc	**4,863 px (Localized)**
Macula / Fovea	**Pos: (169.41, 306.75)**
Retinal Blood Vessels	**11,417 px**
Media Clarity	**Good**

6. RETINAL VISUAL EVIDENCE GALLERY (UNCROPPED FUNDUS PROPORTION)



1. Original Fundus

True color clinical capture



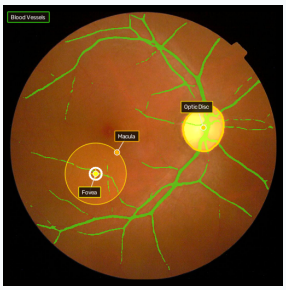
2. Combined Retinal AI

Landmarks & detected lesions



3. Grad-CAM++ Attention

Neural activation heatmap



4. Optic Disc & Vessels

Anatomical reference landmarks

7. AI-DETECTED FINDINGS — DOCTOR-FRIENDLY CLINICAL EXPLANATIONS

Microaneurysm — Detected: 31 px

What it is: A small localized outpouching of a retinal capillary wall where weakened pericytes allow dilation.

Why it matters: Microaneurysms represent the earliest observable retinal microvascular finding associated with diabetic retinopathy.

Clinical Note: The finding should be interpreted together with other retinal abnormalities and the patient's clinical glycemic history.

Retinal Hemorrhage — Detected: 33 px

What it is: Bleeding caused by leakage from damaged and ruptured retinal microvessels into intraretinal layers.

Why it matters: Retinal hemorrhages can occur in diabetic retinopathy and other retinal vascular disorders, indicating microvascular compromise.

Clinical Note: Requires clinical correlation and ophthalmologist assessment to evaluate the extent of vascular compromise.

Hard Exudate — Not Detected (0 px)

What it is: Lipid and lipoprotein deposits precipitated in the retina from abnormal vascular permeability and serum leakage.

Why it matters: They indicate chronic retinal vascular leakage and are important when assessing diabetic retinal disease and proximity to the fovea.

Clinical Note: Evaluate spatial relationship to the foveal avascular zone to monitor risk of diabetic macular edema (DME).

Soft Exudate (Cotton-Wool Spot) — Not Detected (0 px)

What it is: A localized area of retinal nerve-fiber-layer ischemic injury with axoplasmic flow stasis.

Why it matters: It represents focal retinal ischemic change and can occur in diabetic retinopathy as well as other systemic microvascular conditions.

Clinical Note: Reflects localized precapillary arteriolar occlusion and localized retinal ischemia rather than true exudation.

8. CLINICAL INTERPRETATION & DOCTOR'S TRIAGE SUMMARY

Retinal screening examination reveals healthy, well-perfused retinal microvasculature with sharp foveal avascular zone. No microaneurysms, blot hemorrhages, or exudates observed across the posterior pole. Recommendation: Continue routine annual dilated fundus surveillance.

What this means for the patient: The blood vessels at the back of your eye look clear and undamaged today. Keeping your blood sugar and blood pressure within target limits will help keep your vision sharp.

9. REFERRAL RECOMMENDATION & STATUS

Referral Status:	Priority Level:	Target Facility:
Not referred	**Routine**	**Primary Health Centre**

Referral Reason: **No active sight-threatening lesions observed. Routine 12-month follow-up.**

10. SPECIALIST REVIEW & ASSESSMENT

Specialist Review: Pending

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Specialist Assessment: *Pending specialist clinical evaluation.*

Specialist Notes: *This screening record is queued for tele-ophthalmology verification.*

Specialist Recommendation: *Pending clinical review by assigned specialist.*

11. FOLLOW-UP RECOMMENDATION

Routine 12-month follow-up screening recommended.

12. OFFICIAL REGULATORY DISCLAIMER

TRINAY provides AI-assisted screening support and does not replace diagnosis by a qualified eye-care professional. Definitive clinical decisions should always be confirmed through direct dilated slit-lamp biomicroscopy or OCT by an ophthalmologist.